BUSINESS REQUIREMENTS DOCUMENT

UPI Payment Dispute Resolution System

End-to-End Transaction Lifecycle, Failure Handling & Dispute Management

Prepared by: Abhishek Chebolu | Technical Business Analyst | 2025

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DOCUMENT PURPOSE

This BRD defines the functional and non-functional requirements for a UPI Payment Dispute Resolution System. It covers the complete UPI transaction lifecycle, state machine modeling, failure scenarios, idempotency handling,
dispute management workflows, and UAT strategy. This document serves as the single source of truth for all stakeholder alignment, development specification, and UAT sign-off.

1. EXECUTIVE SUMMARY

UPI (Unified Payments Interface) is India's real-time payment system managed by NPCI (National Payments Corporation of India). It enables instant fund transfers between bank accounts through mobile devices using Virtual Payment Addresses (VPA).

Despite high success rates, UPI transactions fail due to network issues, bank timeouts, insufficient balances, and system errors. Failed or disputed transactions require a structured dispute resolution mechanism to ensure payer protection, regulatory compliance, and customer trust.

This system provides a complete end-to-end dispute management platform covering: transaction lifecycle tracking, automated failure detection, dispute initiation, investigation workflow, resolution processing, and refund management.

Metric	Value
UPI Daily Transaction Volume (2024)	~500 million transactions/day
Average Failure Rate	1.5 – 3% of total transactions
Dispute Resolution TAT (RBI mandate)	Within 5 business days
Auto-reversal Trigger	Within 24 hours of failed transaction
Refund Processing Time	T+1 to T+5 depending on bank

2. SCOPE

2.1 In Scope

- UPI payment initiation — collect flow and pay flow
- Transaction state lifecycle management (INITIATED → PENDING → SUCCESS / FAILED / REVERSED)
- Automated failure detection and timeout handling
- Idempotency enforcement to prevent duplicate debits on retry
- Dispute initiation by customer or merchant
- Dispute investigation workflow (Payer PSP → Beneficiary PSP → Bank → NPCI)
- Resolution processing: refund, chargeback, or rejection
- Notification system for all state changes
- Reporting dashboard for operations team

2.2 Out of Scope

- UPI Lite (offline payment functionality)
- International cross-border UPI transactions
- Credit-based UPI products (UPI Credit Line)
- Core banking system (CBS) internal processing
- NPCI internal switch logic

3. STAKEHOLDERS

Stakeholder	Role	Interest / Responsibility
Customer (Payer)	End User	Initiates payment, raises dispute, receives refund
Merchant (Payee)	End User / Beneficiary	Receives payment, responds to dispute claims
Payer PSP	Payment Service Provider	Manages payer's UPI app, routes to NPCI, handles disputes
Beneficiary PSP	Payment Service Provider	Manages payee's VPA, coordinates with beneficiary bank

Stakeholder	Role	Interest / Responsibility
NPCI	Regulator / Switch	Routes transactions, manages UPI switch, dispute arbitration
Payer Bank	Financial Institution	Executes debit, responds to reversal requests
Beneficiary Bank	Financial Institution	Executes credit, responds to chargeback requests
Operations / L2 Team	Internal Team	Investigates escalated disputes, approves refunds
Compliance Team	Internal Team	Ensures RBI / NPCI regulatory adherence
Product Manager	Internal Stakeholder	Owns product decisions, prioritizes features
Technical BA (Author)	Requirements Owner	Documents requirements, facilitates workshops, UAT sign-off

4. UPI PAYMENT FLOW

4.1 Pay Flow (Push Payment)

Customer initiates payment directly from their UPI app to the beneficiary's VPA.

PAYMENT FLOW SEQUENCE

Step 1: Customer enters Beneficiary VPA and amount in UPI app
Step 2: Customer authenticates via MPIN
Step 3: Payer PSP validates VPA and sends Pay Request to NPCI
Step 4: NPCI routes request to Beneficiary PSP
Step 5: Beneficiary PSP validates VPA and confirms account details
Step 6: NPCI sends Debit Request to Payer Bank
Step 7: Payer Bank executes debit and returns Debit Response
Step 8: NPCI sends Credit Request to Beneficiary Bank
Step 9: Beneficiary Bank executes credit and returns Credit Response
Step 10: NPCI sends final status to both PSPs
Step 11: Both PSPs notify respective customers

4.2 Collect Flow (Pull Payment)

Merchant or payee sends a collect request to the customer who approves or rejects it.

COLLECT FLOW SEQUENCE

Step 1: Merchant PSP sends Collect Request to NPCI with customer VPA
Step 2: NPCI routes collect request to Payer PSP
Step 3: Customer receives notification — approves or declines
Step 4: On approval, customer authenticates via MPIN
Step 5: Flow continues identical to Pay Flow from Step 6 onward
Step 6: On decline — transaction marked DECLINED, no fund movement

5. TRANSACTION STATE MACHINE

Every UPI transaction passes through defined states. Understanding these states is critical for dispute resolution, as the state at time of dispute determines the resolution path.

State	Description	Possible Next States
INITIATED	Customer has submitted payment. Request received by Payer PSP. No fund movement yet.	PENDING, FAILED (validation error)
PENDING	Request sent to NPCI. Awaiting bank responses. Outcome unknown — do NOT assume failed.	SUCCESS, FAILED, TIMED_OUT
SUCCESS	Debit and credit both confirmed. Funds transferred. Transaction complete.	DISPUTED (if customer raises dispute)
FAILED	Transaction rejected. Reasons: invalid VPA, insufficient balance, bank rejection, wrong MPIN.	No further state (terminal)
TIMED_OUT	Response not received within defined window. System must trigger status check — NOT auto-failed.	PENDING (on check), SUCCESS, REVERSED
REVERSED	Debit occurred but credit failed. NPCI triggers reversal to return funds to payer.	REVERSAL_PENDING → REVERSAL_SUCCESS
DISPUTED	Customer has raised a formal dispute against a completed transaction.	UNDER_INVESTIGATION, RESOLVED, REJECTED
RESOLVED	Dispute resolved. Refund processed or transaction confirmed as valid.	Terminal state

5.1 Critical State Scenarios

SCENARIO 1: Bank Timeout → PENDING → Late SUCCESS

Trigger: Bank does not respond within timeout window (typically 30 seconds)
State: Transaction moves to TIMED_OUT then PENDING
Action: PSP initiates status check API call after 60-second interval
Outcome A: If debit confirmed and credit confirmed → state moves to SUCCESS
Outcome B: If debit confirmed but credit failed → state moves to REVERSED
Outcome C: If neither confirmed → state moves to FAILED
Risk: System must NOT allow retry debit while original is PENDING — idempotency key prevents double debit

SCENARIO 2: Debit SUCCESS + Credit FAIL → REVERSAL

Trigger: Payer bank confirms debit but beneficiary bank rejects credit
Reason: Beneficiary account frozen, VPA resolved incorrectly, bank system down
State: NPCI moves transaction to REVERSAL_PENDING
Action: NPCI sends reversal request to Payer Bank to return funds
Timeline: Reversal must complete within T+1 per NPCI guidelines
Customer: Receives notification: 'Transaction reversed. Amount will reflect in T+1 day.'
If reversal fails: Escalated to NPCI dispute queue automatically

SCENARIO 3: Duplicate Transaction (Idempotency Enforcement)

Trigger: Customer taps pay twice, or network retry resends same request
Protection: Each payment request carries a unique Idempotency Key (UUID format)
Behavior: If same Idempotency Key received twice, system returns cached response
Result: Second request does NOT create a new debit — prevents double payment
Example: TXN-20250101-ABC123 received twice → only first processed, second returns same SUCCESS response
BA Note: Idempotency Key must be documented as REQUIRED field in all payment API contracts

6. FUNCTIONAL REQUIREMENTS

6.1 Payment Initiation

Req ID	Requirement	Priority	Notes
FR-001	System shall validate VPA format before sending collect/pay request to NPCI	HIGH	Format: name@bankcode
FR-002	System shall authenticate customer via MPIN before processing any fund movement	HIGH	MPIN stored as hash, never plaintext
FR-003	System shall generate a unique Idempotency Key for every new payment request	HIGH	UUID v4 format, 36 characters
FR-004	System shall reject duplicate requests carrying the same Idempotency Key	HIGH	Return 200 with original response
FR-005	System shall enforce per-transaction limit of Rs 1,00,000 per RBI guidelines	HIGH	Configurable by bank/PSP
FR-006	System shall timeout and set state to TIMED_OUT if no bank response in 30s	HIGH	Trigger status check after 60s

6.2 Dispute Management

Req ID	Requirement	Priority	Notes
FR-010	Customer shall be able to raise a dispute within 30 days of transaction date	HIGH	Post 30 days, dispute auto-rejected
FR-011	System shall accept dispute reasons: Amount debited but not credited, Duplicate debit, Merchant non-delivery, Unauthorized transaction	HIGH	Dropdown selection in UI
FR-012	System shall auto-check transaction status before accepting dispute to prevent false disputes on pending transactions	HIGH	If still PENDING, show wait message
FR-013	Payer PSP shall notify Beneficiary PSP within 24 hours of dispute raised	HIGH	Via NPCI dispute channel
FR-014	Operations team shall be able to view, assign, and update dispute status in dashboard	MEDIUM	Role-based access required
FR-015	System shall auto-escalate unresolved disputes to NPCI after 3 business days	HIGH	Per RBI TAT mandate
FR-016	System shall process refund to payer within 5 business days of dispute resolution	HIGH	RBI compliance requirement

6.3 Notification Requirements

Trigger Event	Recipient	Channel	Content
Payment INITIATED	Payer	Push + SMS	Payment of Rs X sent to VPA@bank. TxnID: XXXX
Payment SUCCESS	Both	Push + SMS	Payment successful. Ref: XXXX
Payment FAILED	Payer	Push + SMS	Payment failed. Reason: [code]. Amount not debited.
Transaction REVERSED	Payer	Push + SMS	Amount reversed. Credit in T+1 day.
Dispute RAISED	Both + Ops	Push + Email	Dispute #DIS-XXXX raised. Under review.

Trigger Event	Recipient	Channel	Content
Dispute RESOLVED (Refund)	Payer	Push + SMS	Dispute resolved. Refund of Rs X in 2-3 days.
Dispute REJECTED	Payer	Push + Email	Dispute rejected. Reason: [details]. Contact NPCI.

7. NON-FUNCTIONAL REQUIREMENTS

Category	Requirement	Target
Performance	Payment API response time (P95)	< 2 seconds
Performance	Status check API response time	< 1 second
Availability	System uptime SLA	99.95% (< 4.5 hrs downtime/year)
Scalability	Peak concurrent transactions supported	1 million TPS
Security	MPIN storage	Hashed (SHA-256), never logged
Security	API communication encryption	TLS 1.2 minimum
Compliance	RBI dispute resolution TAT	5 business days
Compliance	Data retention for transaction records	5 years per RBI mandate
Idempotency	Duplicate key detection window	24 hours
Auditability	All state transitions must be logged	With timestamp + actor ID

8. API CONTRACTS

All APIs follow REST architecture. Base URL: https://api.upidispute.com/v1

API 1: Initiate Payment

Field	Value
Method	POST
Endpoint	/payments/initiate
Auth	Bearer Token (OAuth 2.0)
Idempotency	Header: X-Idempotency-Key (Required, UUID v4)

Request Field	Type	Required	Description
payer_vpa	String	Yes	Payer's Virtual Payment Address (e.g., abhi@okaxis)
payee_vpa	String	Yes	Beneficiary's VPA
amount	Decimal	Yes	Transaction amount in INR (max: 100000.00)
remarks	String	No	Payment note/description (max 50 chars)
device_id	String	Yes	Unique device identifier for fraud check
idempotency_key	String	Yes	UUID v4 — prevents duplicate processing

Response Field	Type	Description
transaction_id	String	Unique TxnID assigned by system (e.g., TXN20250101ABC123)
status	String	INITIATED / PENDING / SUCCESS / FAILED
timestamp	DateTime	ISO 8601 format (e.g., 2025-01-01T10:30:00Z)
message	String	Human-readable status description

Error Code	HTTP Status	Meaning
PAY_001	400	Invalid VPA format — VPA does not match name@bankcode pattern
PAY_002	400	Amount exceeds limit — transaction exceeds Rs 1,00,000
PAY_003	401	MPIN authentication failed — incorrect MPIN entered
PAY_004	402	Insufficient balance — payer account has insufficient funds
PAY_005	409	Duplicate idempotency key — same key used within 24 hours
PAY_006	503	Bank unavailable — payer/beneficiary bank system is down
PAY_007	504	Gateway timeout — no response from NPCI within timeout window

API 2: Check Transaction Status

Field	Value
Method	GET
Endpoint	/payments/{transaction_id}/status
Auth	Bearer Token

Response Field	Type	Description
transaction_id	String	Original transaction ID
status	String	Current state: INITIATED / PENDING / SUCCESS / FAILED / REVERSED
debit_status	String	Payer bank debit confirmation: CONFIRMED / PENDING / FAILED
credit_status	String	Beneficiary bank credit confirmation: CONFIRMED / PENDING / FAILED
last_updated	DateTime	Last state change timestamp
reversal_ref	String	Populated only if status is REVERSED

API 3: Raise Dispute

Field	Value
Method	POST
Endpoint	/disputes/raise
Auth	Bearer Token

Request Field	Type	Required	Description
transaction_id	String	Yes	ID of the disputed transaction

Request Field	Type	Required	Description
dispute_reason	String	Yes	Enum: AMOUNT_NOT_CREDITED / DUPLICATE_DEBIT / UNAUTHORIZED / NON_DELIVERY
description	String	No	Customer's description of issue (max 200 chars)
contact_number	String	Yes	Customer contact for follow-up

Response Field	Type	Description
dispute_id	String	Unique dispute reference (e.g., DIS-20250101-XYZ)
status	String	RAISED — initial state on creation
expected_resolution_date	Date	T+5 business days from raise date

API 4: Check Dispute Status

Field	Value
Method	GET
Endpoint	/disputes/{dispute_id}/status
Auth	Bearer Token

Response Field	Type	Description
dispute_id	String	Dispute reference ID
status	String	RAISED / UNDER_INVESTIGATION / RESOLVED / REJECTED / ESCALATED
assigned_to	String	Ops team member ID handling the dispute
resolution_remarks	String	Populated when status is RESOLVED or REJECTED
refund_ref	String	Refund transaction ID if refund initiated

API 5: Process Refund

Field	Value
Method	POST
Endpoint	/disputes/{dispute_id}/refund
Auth	Bearer Token (Ops role)

Request Field	Type	Required	Description
refund_amount	Decimal	Yes	Amount to refund (must be <= original transaction amount)
refund_type	String	Yes	FULL / PARTIAL
ops_agent_id	String	Yes	Operations agent approving the refund
resolution_notes	String	Yes	Reason for refund approval (internal log)

9. USER STORIES

Format: As a [persona], I want to [action], so that [benefit]. Acceptance Criteria written in Given-When-Then format.

Story ID	Persona	User Story	Priority
US-001	Customer	As a customer, I want to initiate a UPI payment using a VPA, so that I can transfer funds instantly without bank details.	HIGH
US-002	Customer	As a customer, I want to receive a real-time notification on payment success or failure, so that I have immediate confirmation.	HIGH
US-003	Customer	As a customer, I want my money to be auto-reversed within T+1 if credit fails after debit, so that I don't lose funds.	HIGH
US-004	Customer	As a customer, I want to raise a dispute for an unresolved failed payment, so that I can claim my refund through a formal process.	HIGH
US-005	Customer	As a customer, I want to track the status of my dispute in real time, so that I don't need to call customer care.	HIGH
US-006	Customer	As a customer, I want to receive my refund within 5 business days of dispute resolution, so that my funds are returned quickly.	HIGH
US-007	Customer	As a customer, I want the system to prevent duplicate debits on payment retry, so that I am never charged twice.	HIGH
US-008	Merchant	As a merchant, I want to initiate a collect request to my customer's VPA, so that I can request payment without sharing bank details.	HIGH
US-009	Merchant	As a merchant, I want to receive notification when a collect request is approved or rejected, so that I can manage my order workflow.	MEDIUM
US-010	Merchant	As a merchant, I want to respond to a dispute raised against my transaction, so that I can provide evidence for a valid transaction.	HIGH
US-011	Ops Team	As an operations agent, I want to view all open disputes in a dashboard, so that I can prioritize and resolve them within SLA.	HIGH
US-012	Ops Team	As an operations agent, I want to update the dispute status and add resolution notes, so that the customer is kept informed.	HIGH
US-013	Ops Team	As an operations agent, I want to process a full or partial refund against a resolved dispute, so that funds are returned to the customer.	HIGH
US-014	Ops Team	As an operations agent, I want disputes to auto-escalate to NPCI after 3 days if unresolved, so that SLA breach is prevented.	HIGH
US-015	System	As the system, I want to enforce idempotency on all payment requests using unique keys, so that duplicate payments are automatically blocked.	HIGH
US-016	System	As the system, I want to automatically trigger a status check API when a transaction is in TIMED_OUT state, so that the correct final state is determined.	HIGH
US-017	System	As the system, I want to log every transaction state change with timestamp and actor, so that a complete audit trail is available.	MEDIUM
US-018	Customer	As a customer, I want to select from predefined dispute reasons, so that my dispute is categorized correctly for faster resolution.	MEDIUM
US-019	Ops Team	As an operations manager, I want a daily report of disputes by status and TAT, so that I can monitor team performance and compliance.	MEDIUM
US-020	Customer	As a customer, I want to be notified when my dispute is escalated to NPCI, so that I know the investigation is in progress.	MEDIUM

9.1 Acceptance Criteria — Sample Stories

> **US-003: Auto-Reversal Acceptance Criteria**

Given: A UPI payment where debit was confirmed by payer bank
When: Credit fails at beneficiary bank (rejected or bank down)
Then: System shall automatically initiate reversal request to payer bank within 1 hour
And: Transaction state shall move from FAILED to REVERSAL_PENDING
And: Customer shall receive push notification: 'Amount reversed, credit in T+1'
And: If reversal fails, dispute shall be auto-raised and escalated to NPCI

US-007: Idempotency Acceptance Criteria

Given: A payment request with Idempotency-Key: TXN-UUID-123 was processed successfully
When: Same request with Idempotency-Key: TXN-UUID-123 is received again within 24 hours
Then: System shall NOT process a new debit
And: System shall return HTTP 200 with the original transaction response
And: Original transaction ID shall be returned in response body
And: Duplicate attempt shall be logged in audit trail with flag: DUPLICATE_BLOCKED

10. UAT TEST CASES

TC ID	Scenario	Precondition	Steps	Expected Result	Priority
TC-001	Successful UPI payment — happy path	Valid VPA, sufficient balance, bank online	Enter VPA, amount, MPIN → Submit	Status: SUCCESS, both parties notified, funds transferred	HIGH
TC-002	Invalid VPA format	None	Enter VPA without @bankcode, submit	Error PAY_001: Invalid VPA format. Payment not processed.	HIGH
TC-003	Insufficient balance	Payer balance < payment amount	Valid VPA, amount > balance, submit	Error PAY_004: Insufficient balance. State: FAILED.	HIGH
TC-004	Wrong MPIN — 3 attempts	Valid VPA, sufficient balance	Enter wrong MPIN 3 times	Account locked after 3 failures. Error PAY_003. Notification sent.	HIGH
TC-005	Bank timeout — PENDING state	Simulate bank non-response	Submit payment → bank timeout at 30s	State: TIMED_OUT → PENDING. Status check triggered at 60s.	HIGH
TC-006	Debit success + credit fail → REVERSAL	Beneficiary bank returns error after debit	Simulate credit rejection post-debit	State: REVERSED. Payer notified. Reversal ref created.	HIGH
TC-007	Duplicate payment — idempotency	Successful TXN with Idempotency Key AAA	Resend same request with Key AAA	HTTP 200 returned with original response. No new debit.	HIGH
TC-008	Duplicate payment — different key	None	Same amount to same VPA with new key	New TXN processed. Both transactions visible in history.	MEDIUM
TC-009	Raise dispute — amount not credited	Transaction status: SUCCESS, customer claims no credit	Customer raises dispute, selects AMOUNT_NOT_CREDITED	Dispute ID created. Status: RAISED. Ops team notified.	HIGH

TC ID	Scenario	Precondition	Steps	Expected Result	Priority
TC-010	Raise dispute — outside 30 days	Transaction older than 30 days	Customer attempts to raise dispute	Error: Dispute window expired. Redirect to NPCI grievance portal.	HIGH
TC-011	Dispute status tracking	Dispute DIS-001 exists in UNDER_INVESTIGATION	Customer checks /disputes/DIS-001/status	Returns: status=UNDER_INVESTIGATION, assigned_to, expected_resolution_date	MEDIUM
TC-012	Ops agent resolves dispute — full refund	Dispute DIS-001 in UNDER_INVESTIGATION	Ops agent approves full refund via dashboard	Status: RESOLVED. Refund TXN created. Customer notified.	HIGH
TC-013	Ops agent rejects dispute	Dispute DIS-002 — investigation shows valid TXN	Ops agent marks REJECTED with reason	Status: REJECTED. Customer notified with reason. Audit logged.	HIGH
TC-014	Auto-escalation to NPCI after 3 days	Dispute DIS-003 unresolved for 3 business days	Wait for 3-day threshold / simulate in test env	Status: ESCALATED. NPCI notified. Customer informed.	HIGH
TC-015	Transaction limit enforcement	None	Submit payment for Rs 1,00,001	Error PAY_002: Amount exceeds limit. Payment blocked.	HIGH
TC-016	Collect request — customer approves	Merchant initiates collect request	Customer receives and approves collect request + MPIN	Payment processed identically to pay flow. Both notified.	HIGH
TC-017	Collect request — customer declines	Merchant initiates collect request	Customer declines collect request	Status: DECLINED. No fund movement. Merchant notified.	MEDIUM
TC-018	Partial refund on resolved dispute	Dispute for Rs 500 — partial resolution approved	Ops processes partial refund of Rs 300	Refund TXN for Rs 300 created. Customer notified of partial.	MEDIUM
TC-019	Audit trail completeness	Any transaction processed	Admin checks audit log for TXN-001	All state transitions logged with timestamp, actor, and reason.	MEDIUM
TC-020	Concurrent payment requests — idempotency under load	Simulate 10 concurrent retries same Idempotency Key	Fire 10 requests simultaneously with same key	Only 1 debit processed. 9 return cached 200 response.	HIGH
TC-021	Status check on PENDING transaction	Transaction in PENDING state	Call /payments/{id}/status	Returns debit_status, credit_status individually. State: PENDING.	HIGH
TC-022	Refund credit to correct account	Dispute resolved, refund initiated	Monitor refund TXN	Refund credited to original payer VPA only. No misdirection.	HIGH
TC-	Notification delivery —	Payment fails with PAY_004	Trigger failed payment	Push notification AND SMS delivered to payer within 5 seconds.	MEDIUM

TC ID	Scenario	Precondition	Steps	Expected Result	Priority
023	payment failure				
TC-024	Dispute raised on PENDING transaction	Transaction in PENDING state	Customer attempts to raise dispute	System shows: 'Transaction still being processed. Retry after 24 hours.'	MEDIUM
TC-025	Session expiry during payment	User idle for > 5 minutes during payment	Start payment, idle 6 minutes, submit	Session expired error. Transaction NOT initiated. No fund movement.	MEDIUM

11. DATA DICTIONARY

Field Name	Data Type	Format / Length	Description	Example
transaction_id	String	TXN+YYYYMMDD+6chars, 17 chars	Unique transaction identifier	TXN20250101ABC123
payer_vpa	String	name@bankcode, max 50 chars	Payer Virtual Payment Address	abhi@okaxis
payee_vpa	String	name@bankcode, max 50 chars	Beneficiary VPA	merchant@ybl
amount	Decimal	Max 10 digits, 2 decimal	Transaction amount in INR	5000.00
status	Enum	Max 20 chars	Current transaction state	PENDING
idempotency_key	String	UUID v4, 36 chars	Unique key to prevent duplicates	123e4567-e89b-12d3
dispute_id	String	DIS-YYYYMMDD-6chars	Unique dispute reference	DIS-20250101-XYZ001
dispute_reason	Enum	Max 30 chars	Category of dispute	AMOUNT_NOT_CREDITED
debit_status	Enum	CONFIRMED/PENDING/FAILED	Payer bank debit confirmation	CONFIRMED
credit_status	Enum	CONFIRMED/PENDING/FAILED	Beneficiary bank credit status	FAILED
reversal_ref	String	REV+TXN_ID format	Reversal transaction reference	REVTXN20250101ABC123
refund_amount	Decimal	Max 10 digits, 2 decimal	Refund amount processed	5000.00
refund_type	Enum	FULL / PARTIAL	Type of refund	FULL
timestamp	DateTime	ISO 8601	Event timestamp	2025-01-01T10:30:00Z
device_id	String	Max 64 chars	Device identifier for fraud	DEV-A1B2C3D4

12. ASSUMPTIONS, CONSTRAINTS & SIGN-OFF

12.1 Assumptions

- NPCI UPI switch and all PSP integrations are available and compliant with NPCI technical specifications.
- Customer VPA validation is performed by the PSP before sending to NPCI — invalid VPAs are rejected at PSP level.
- All banks participating in UPI maintain 99.9%+ uptime — downtime is treated as an exception scenario.
- Idempotency key is generated client-side (PSP app) and is guaranteed unique per device session.
- Audit logs are retained for minimum 5 years per RBI data retention guidelines.

12.2 Constraints

- Transaction limit: Rs 1,00,000 per transaction (configurable per bank, max Rs 2,00,000 for some categories).
- Dispute window: 30 days from transaction date — no exceptions without NPCI escalation.
- Dispute resolution TAT: 5 business days per RBI mandate — non-negotiable.
- All payment data must be stored within India per RBI data localization guidelines.

12.3 Document Sign-Off

Role	Name	Signature	Date
Technical Business Analyst	Abhishek Chebolu	_____	_____
Product Manager	_____	_____	_____
Engineering Lead	_____	_____	_____
QA Lead	_____	_____	_____
Compliance Officer	_____	_____	_____

PREPARED BY
Abhishek Chebolu | Technical Business Analyst
Email: abhichebolu@gmail.com | Location: Hyderabad, India
GitHub: [Your GitHub URL] | LinkedIn: [Your LinkedIn URL]
This document is part of a BA portfolio project demonstrating applied analysis of BFSI / Fintech systems.